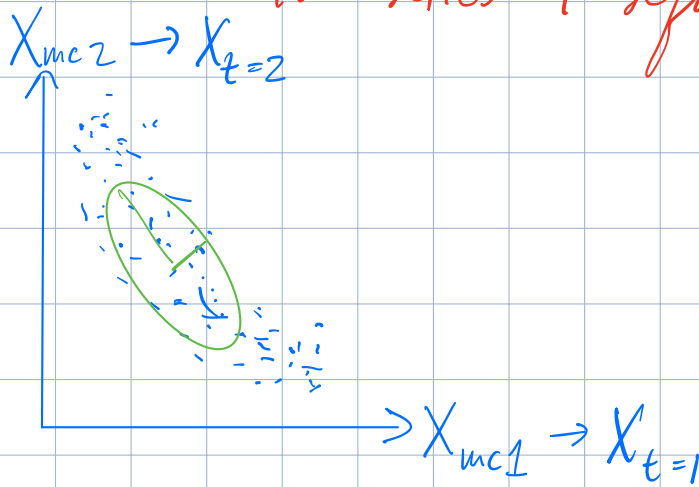


Time Series & Sequential Decisions



UNIT 1

Stochastic Processes

Let us stack all the instances of the stochastic process into a $\bar{t}$ -dimensional vector

$$\mathbf{X}_{[\bar{t}]} = \begin{pmatrix} X_1 \\ \vdots \\ X_{\bar{t}} \end{pmatrix}. \quad (41.2)$$

Such a random vector admits a probabilistic representation in terms of its cdf (2.9) (or equivalently pdf, characteristic function etc. (2.7))

$$\mathbf{X}_{[\bar{t}]} \sim f(x_1, \dots, x_{\bar{t}}). \quad (41.3)$$

On the other hand, if each of the X_t has finite variance, we can limit ourselves to represent randomness as described the mean-covariance representation (6.26)

$$\mathbf{X}_{[\bar{t}]} \stackrel{mc}{\sim} \{\mu([\bar{t}]), k([\bar{t}], [\bar{t}])\}, \quad (41.4)$$

where

- $\mu_X([\bar{t}])$ is the vector of the respective expectations (6.14)

$$\mu([\bar{t}]) \equiv \begin{pmatrix} \mathbb{E}\{X_1\} \\ \vdots \\ \mathbb{E}\{X_{\bar{t}}\} \end{pmatrix}; \quad (41.5)$$

- $k_X([\bar{t}], [\bar{t}])$ is the matrix of the respective cross-covariances (6.17)

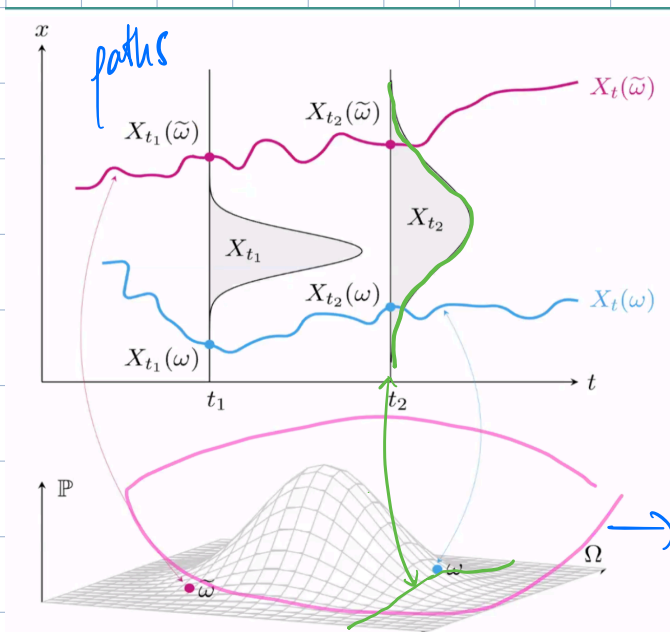
$$k([\bar{t}], [\bar{t}]) \equiv \begin{pmatrix} \mathbb{V}\{X_1\} & \cdots & \text{Cov}\{X_1, X_{\bar{t}}\} \\ \vdots & \ddots & \vdots \\ \text{Cov}\{X_{\bar{t}}, X_1\} & \cdots & \mathbb{V}\{X_{\bar{t}}\} \end{pmatrix}. \quad (41.6)$$

Stacking RVs:

In full generality, a **stochastic process** $\{\mathbf{X}_t\}_{t \in \mathcal{T}}$ is a collection of random variables (53.10) on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ (53.1)

$$\{\mathbf{X}_t \equiv \begin{pmatrix} X_{1,t} \\ \vdots \\ X_{\bar{n},t} \end{pmatrix}\}_{t \in \mathcal{T}} \quad \text{where} \quad \mathbf{X}_t(\cdot) : \begin{matrix} \Omega \rightarrow \mathbb{R}^{\bar{n}} \\ \omega \mapsto \mathbf{X}_t(\omega) \end{matrix},$$

where $\mathcal{T} = \mathbb{Z}$ for continuous & $\mathcal{T} = \mathbb{R}$ for discrete



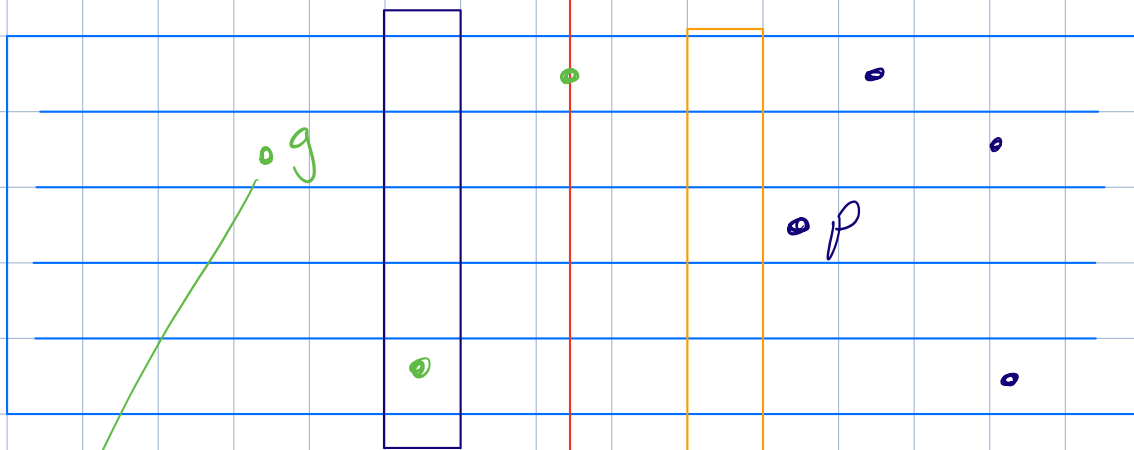
→ sequences where paths have $\mathbb{P}$

→ sample abstract space

in Excel: $X_{m,t}$

realized ← now → future

m
— instrument ↑



each is an RV

simultaneous

time →
t

$$X_P \sim f_X(x_{n_1 t_1}, \dots, x_{n_k t_k}) \rightarrow \text{joint pdf}$$

For a given multivariate stochastic process $\{X_t \equiv (X_{1,t}, \dots, X_{n,t})'\}_{t \in T}$ (41.1), consider an arbitrary joint set of an arbitrary finite number k of joint cross-sectional and time indices P (41.16), and the ensuing random vector X_P (41.17).

Then we can specify the mean-covariance class for the random vector X_P (41.17) in terms of its mean vector and covariance matrix

$$X_P \stackrel{mc}{\sim} \{\mu_X(P), k_X(P, P)\}.$$

(41.35)

where

- We overload notation for the synchronous expectation function (41.30) with the (non-synchronous) **mean function** which takes as input any finite set of double indices P (41.16) of cardinality k and outputs the k -dimensional vector of the expectation of the instances of the process

$$\mu_X(P) \equiv \mathbb{E}\{X_P\} = \begin{pmatrix} \mathbb{E}\{X_{n_1, t_1}\} \\ \vdots \\ \mathbb{E}\{X_{n_k, t_k}\} \end{pmatrix}.$$

(41.36)

- We overload the notation for the autocovariance function (41.31) with the **autocovariance kernel** which takes as inputs any two finite set of indices $\mathcal{P}$ of cardinality k (41.16) and $\mathcal{G}$ of cardinality j (41.22), and outputs the $k \times j$ matrix of the cross-covariances (6.21) of the instances of the process

$$k_X(\mathcal{P}, \mathcal{G}) \equiv \text{Cv}\{X_{\mathcal{P}}, X_{\mathcal{G}}\} \equiv \begin{pmatrix} \text{Cv}\{X_{n_1, t_1}, X_{m_1, s_1}\} & \cdots & \text{Cv}\{X_{n_1, t_1}, X_{m_j, s_j}\} \\ \vdots & \ddots & \vdots \\ \text{Cv}\{X_{n_k, t_k}, X_{m_1, s_1}\} & \cdots & \text{Cv}\{X_{n_k, t_k}, X_{m_j, s_j}\} \end{pmatrix}. \quad (41.39)$$

• in MCF, we only need the ellipsoid for a pair instead of a joint PDF

§ slices above, specify mean function

$\mu_x(t) = \mathbb{E}(X_t)$ § autocovariance for 2

slices, $k_x(\mu, t) \equiv \text{Cv}\{X_\mu, X_t\}$

↳ needs to be positive definite
called a "kernel"

how to make a prediction?
probabilistic

If the stochastic process X_t (41.7) is specified as probabilistic process (41.11), then prediction is performed by means of conditioning (3.14)

$$X_{\mathcal{P}} | x_{\mathcal{G}} \sim f(x_{\mathcal{P}} | x_{\mathcal{G}}). \quad (41.55)$$

The conditional distribution follows from the distribution (41.20) for the joint set of variables to be predicted (41.17) and given (41.23), using the marginalization consistency (41.24)

$$f(x_{\mathcal{P}} | x_{\mathcal{G}}) = \frac{f(x_{\mathcal{P} \cup \mathcal{G}})}{\int f(x_{\mathcal{P} \cup \mathcal{G}}) dx_{\mathcal{P}}}, \quad (41.56)$$

where the vector $X_{\mathcal{P} \cup \mathcal{G}}$ (41.23) is meant as the joint vector $(X_{\mathcal{P}}, X_{\mathcal{G}})$.

In general, the computation of the conditional distribution (41.56) is challenging in two respects: we do not have the joint distribution for all indices, and it is hard to compute the integration in the denominator.

prediction in MCF

41.2.2 Kriging

If the stochastic process $\mathbf{X}_t$ (41.65) is specified up to its mean-covariance class (41.28), namely in terms of expectation function $\mu_{\mathbf{X}}(t)$ (41.30) and autocovariance function $k_{\mathbf{X}}(t, u)$ (41.31), then the general prediction (41.55) is performed by means of linear projection (7.43)

$$\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}^{mc} \sim \begin{cases} \mathbb{E}\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\} \\ \mathbb{C}v\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\}. \end{cases} \quad (41.57)$$

$\mathcal{G} = \text{given}$
 $\mathcal{P} = \text{pred}$

Then, the linear prediction $\mathbb{E}\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\}$ (7.27) reads

$$\mathbb{E}\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\} = \hat{\mu}_{\mathbf{X}}(\mathcal{P}) + k_{\mathbf{X}}(\mathcal{P}, \mathcal{G})(k_{\mathbf{X}}(\mathcal{G}, \mathcal{G}))^{-1}(\mathbf{x}_{\mathcal{G}} - \mu_{\mathbf{X}}(\mathcal{G})), \rightarrow \text{just a regression} \quad (41.58)$$

and the prediction error $\mathbb{C}v\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\}$ (7.34) reads

$$\mathbb{C}v\{\mathbf{X}_{\mathcal{P}} \parallel \mathbf{x}_{\mathcal{G}}\} = \hat{k}_{\mathbf{X}}(\mathcal{P}, \mathcal{P}) - k_{\mathbf{X}}(\mathcal{P}, \mathcal{G})(k_{\mathbf{X}}(\mathcal{G}, \mathcal{G}))^{-1}k_{\mathbf{X}}(\mathcal{P}, \mathcal{G})'. \quad (41.59)$$

In the above $\mu_{\mathbf{X}}(\cdot)$ is the mean kernel (41.36); and $k_{\mathbf{X}}(\cdot, \cdot)$ is the autocovariance kernel (41.39), which both are readily computed from the expectation function as in (41.43) and the autocovariance function as in (41.45).

estimate kernels $\hat{k}_{\mathbf{X}}$ in practice, lit. often postulates

how to get to $\hat{k}_{\mathbf{X}}$?
you need:

1 stationarity
- probabilistic distribution (or ellipse in MCF) is invariant by translation

2. ergodicity

Strong ergodicity can be relaxed for covariance stationary processes (41.76), by requiring that the property (41.81) holds true only for a couple of choices of the function h , which are compatible with the solely knowledge of the first two moments.

More precisely, a discrete time (41.8) covariance stationary (41.76) stochastic process $\{\mathbf{X}_t\}_{t \in \mathbb{Z}}$ (41.7) is said to be **ergodic in mean** if its time average converges to the unconditional expectation $\mathbb{E}\{\mathbf{X}_t\} = \mu_{\mathbf{X}}$ (41.73)

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=-t}^t \mathbf{X}_i = \mu_{\mathbf{X}}, \quad (41.82)$$

where the limit must be taken at least in probability [W]. Note how the left-hand side of (41.82) does not depend on t because of covariance stationarity (41.76).

An ergodic in mean process (41.82) is **ergodic in autocovariance** if the time average of the autocovariances $\mathbb{C}v\{\mathbf{X}_{t+\tau}, \mathbf{X}_t\}$ (6.21) converges, in probability, to the true autocovariance $\kappa_{\mathbf{X}}(\tau)$ (41.74)

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=-t}^t (\mathbf{X}_i - \mu_{\mathbf{X}})(\mathbf{X}_{i+\tau} - \mu_{\mathbf{X}})' = \kappa_{\mathbf{X}}(\tau) \quad (41.83)$$

special instance: $X_{m,t}$ from above $\sim$ iid

$$\varepsilon_t \in \text{iid} \sim f(\varepsilon)$$

wlog, $\mathbb{E}\{\varepsilon_t\} = 0 \longrightarrow$ strong white noise

$$\varepsilon_t \sim_{mc} \{0, \hat{\sigma}_n\}$$

$\hookrightarrow n \times n$ matrix
mean is zero

$\longrightarrow$ weak white noise

residuals?

Random walk 42.2

$$X_{t+1} = \mu + \underbrace{X_t}_{\text{def recursively}} + \varepsilon_{t+1} \quad \text{where wlog } \mathbb{E}(\varepsilon_t) = 0$$

More generally, a **multivariate (probabilistic) random walk (with drift)** $\{X_t\}_{t=0,1,\dots}$ is a stochastic process (41.7) such that for $t > 0$ the next step X_{t+1} is determined by the "current" position X_t plus an unpredictable shock, namely a strong white noise (42.2) with distribution F_ε (2.7)

$$X_{t+1} = \mu + X_t + \varepsilon_{t+1}, \quad (42.6)$$

where without loss of generality $\mathbb{E}\{\varepsilon_t\} = 0$. The process at time zero is set to have an arbitrary distribution

$$X_0 \sim f(x_0) \quad (42.7)$$

uncorrelated with the driving strong white noise ε_t at all times.

Note that any linear combination of the random walk (42.6) also follows a random walk (42.6). Namely, for any $\overline{m} \times \overline{n}$ matrix c the process

$$Y_t \equiv cX_t \quad (42.8)$$

is still an $\overline{m}$ -dimensional random walk (43.13)

$$Y_{t+1} = Y_t + \nu + \eta_{t+1}, \quad (42.9)$$

where $\nu \equiv c\mu$ and $\eta \sim c\varepsilon_t$ i.i.d..

In particular, the generic n -th marginal component $X_{n,t}$ (43.13) ensuing from $c \equiv \delta^{(n)}$ (42.9), follows a univariate random walk (42.5) of its own.

The most important property of a random walk (42.6) is that their increments are i.i.d.

$$\Delta X_{t+1} \sim f(\varepsilon) \text{ i.i.d..} \quad (42.10)$$

Then

- i) **all** the finite-dimensional distributions (41.20) of the random walk (42.6) are fully determined from the knowledge of the distribution $f(\varepsilon)$ of the increment and the inception distribution $f(x_0)$ (red);
- ii) the increment process over **any** time interval $[t, t + \Delta t]$ is i.i.d.

$$\varepsilon_{\Delta t} \equiv X_{t+\Delta t} - X_t = \varepsilon_{t+1} + \dots + \varepsilon_{t+\Delta t} \sim f_{\Delta t}(\varepsilon) \text{ i.i.d..} \quad (42.11)$$

The distribution $f_{\Delta t}(\varepsilon)$ (42.11) is computed via convolution Figure 42.1.

In particular, the increments over any pair of non-overlapping time intervals $[s, s + \Delta s], [t, t + \Delta t]$ are independent and their distribution depends only on the interval lengths $\Delta s, \Delta t$.

$$X_{t+1} = X_t + \varepsilon_{t+1} = \varepsilon_{t-1} + \varepsilon_t + \dots$$

$$\text{Var}(X_t) = t \text{Var}(\varepsilon_t) = t + \sigma^2$$

disgression into Brownian motion & poisson processes $\sim N(\mu, \sigma^2)$ due to CLT

stable Levy process w/ Cauchy increments

$$dX_t \sim \text{Cauchy}(\mu dt, (\sigma dt)^2) \text{ iid}$$

no mean $\rightarrow$ yes median

UNIT 2: linear stochastic models

43.2.1 Definitions

In this section we define the autoregressive of order one process in terms of structural representation, namely an autoregressive equation, and kernel representation, namely as equivalence class (41.28) of processes with given expectation (41.30) and autocovariance function (41.31).

The **autoregressive of order one** AR(1) process is a stochastic process, defined over the positive integers $t \in \mathbb{N}$, such that, for $t = 0, 1, 2, \dots$ satisfies the below difference equation

$$X_{t+1} = bX_t + \varepsilon_{t+1}, \quad (43.14)$$

where b is an arbitrary scalar value; ε_t is an univariate weak white noise (43.1) with zero mean $\mu_\varepsilon = 0$ and variance σ_ε^2 . $\hookrightarrow$ not iid, $\mu=0$ & $\sigma^2=c$

More generally, we can replace the zero-mean weak white noise ε_{t+1} with an arbitrary, univariate weak white noise (43.1). Then the AR(1) equation (43.14) becomes (43.15)

$$X_{t+1} = a + bX_t + \varepsilon_{t+1}, \quad (43.15)$$

where a is an arbitrary scalar "shift" parameter, ε_t is an univariate weak white noise (43.1) with zero mean $\mu_\varepsilon = 0$ and variance σ_ε^2 .

The process at time zero is set to have an arbitrary mean-covariance distribution (6.26)

$$X_0 \overset{mc}{\sim} \begin{cases} \mu_0 \equiv \mathbb{E}\{X_0\} \\ \sigma_0^2 \equiv \text{V}\{X_0\}, \end{cases} \quad (43.16)$$

uncorrelated with the driving weak white noise ε_t at all times.

AR(1)

"Random walk" (43.13)

Cov. stationary (41.74)

Weak white noise (43.1)

$$X_{t+1} = a + bX_t + \varepsilon_{t+1}$$

$$b = 1$$

$$|b| < 1, t \in \mathbb{Z}$$

$$b = 0$$

Table 43.3: AR(1) as b varies

vector autoregression, order of 1 $\rightarrow$ multivariate equivalent

VAR(1)

Vector autoregressive of order one process

$$\mathbf{X}_{t+1} = \mathbf{a} + \mathbf{b}\mathbf{X}_t + \hat{\varepsilon}_{t+1} \sim wvn(0, \sigma_{\hat{\varepsilon}}^2)$$

	AR(1)	VAR(1)
"Random walk"	$b = 1$	$\mathbf{b} = \mathbb{I}_{\bar{n}}$
cov. stationary	$ b < 1$	$ eigval(\mathbf{b}) < 1$
Weak white noise	$b = 0$	$\mathbf{b} = \mathbf{0}_{\bar{n} \times \bar{n}}$
Cointegrated	N.A.	$\begin{cases} eigval(\mathbf{b}) \leq 1 \\ eigval(\mathbf{b}) \ni 1 \end{cases}$

Table: VAR(1) vs AR(1)

Covariance stationary VAR(1)

$|eigval(\mathbf{b})| < 1 \Rightarrow$ Covariance stationary VAR(1)

$$\mathbf{X}_{t+1} = \mathbf{a} + \mathbf{b}\mathbf{X}_t + \hat{\varepsilon}_{t+1} \sim wvn(0, \sigma_{\hat{\varepsilon}}^2)$$

- Expectation function

$$\mu_{\mathbf{X}} = (\mathbb{I}_{\bar{n}} - \mathbf{b})^{-1} \mathbf{a}$$

- Autocovariance function ($\tau > 0$)

$$\kappa_{\mathbf{X}}(\tau) = \mathbf{b}^{\tau} \text{vec}^{-1}[(\mathbb{I}_{\bar{n}^2} - \mathbf{b} \otimes \mathbf{b})^{-1} \text{vec}(\sigma_{\hat{\varepsilon}}^2)]$$

exponential decay

only depends on lag between two points
shift invariant $\rightarrow$ A between t & $t+n$ does not matter

Cointegration

$|eigval(\mathbf{b})| \leq 1$
 $\lambda_1 = \dots = \lambda_{\bar{m}} = 1 \Rightarrow$ Cointegrated VAR(1)

$$\mathbf{X}_{t+1} = \mathbf{a} + \mathbf{b}\mathbf{X}_t + \hat{\varepsilon}_{t+1} \sim wvn(0, \sigma_{\hat{\varepsilon}}^2)$$

$\mathbf{X}_t$ not cov. stationary, only linear combinations $\mathbf{c}'\mathbf{X}_t$ cov. stationary

$\Updownarrow$

Error correction representation

$$\Delta \mathbf{X}_{t+1} = \mathbf{a} + \mathbf{d}\mathbf{c}'\mathbf{X}_t + \hat{\varepsilon}_{t+1}$$

$$\mathbf{b} \mathbb{I}_{\bar{n}} = \mathbf{d}\mathbf{c}'$$

$\bar{n} \times (\bar{n} - \bar{m})$ full-rank matrices
not unique

Ornstein-Uhlenbeck process:

$\uparrow$ time, brownian motion converges to asymptotic dist

you want mean reverting portfolios (buy low, wait for mean reversion)

Cov(X) grows ind. unless in one hyperplane

Forecast

$|eigval(\mathbf{b})| < 1 \Rightarrow$ Covariance stationary VAR(1)

$$\mathbf{X}_{t+1} = \mathbf{a} + \mathbf{b}\mathbf{X}_t + \hat{\varepsilon}_{t+1} \sim wvn(0, \sigma_{\hat{\varepsilon}}^2)$$

- Linear forecast

$$\mathbb{E}\{\mathbf{X}_t | \mathbf{x}_0\} = \mathbf{b}^t \mathbf{x}_0 + (\mathbb{I}_{\bar{n}} - \mathbf{b}^t) \underbrace{(\mathbb{I}_{\bar{n}} - \mathbf{b})^{-1} \mathbf{a}}_{\mu_{\mathbf{X}}}$$

- Linear forecast error

$$\text{vec}(\mathbb{C}v\{\mathbf{X}_t | \mathbf{x}_0\}) = (\mathbb{I}_{\bar{n}^2} - (\mathbf{b} \otimes \mathbf{b})^t) \underbrace{(\mathbb{I}_{\bar{n}^2} - \mathbf{b} \otimes \mathbf{b})^{-1} \text{vec}(\sigma_{\hat{\varepsilon}}^2)}_{\sigma_{\mathbf{X}}^2}$$

$t \gg 0 \Rightarrow$ forecast reverts to unconditional moments

$$\mathbb{E}\{\mathbf{X}_t | \mathbf{x}_0\} \xrightarrow{t \rightarrow \pm \infty} \mu_{\mathbf{X}} \quad \mathbb{C}v\{\mathbf{X}_t | \mathbf{x}_0\} \xrightarrow{t \rightarrow \pm \infty} \sigma_{\mathbf{X}}^2$$

Forecast

Vector autoregressive of order one process

$$\mathbf{X}_{t+1} = \mathbf{a} + \mathbf{b}\mathbf{X}_t + \hat{\varepsilon}_{t+1} \rightarrow \sim wvn(\mathbf{0}, \sigma_{\hat{\varepsilon}_t}^2)$$

- Linear forecast if $\mathbf{b} \approx \mathbb{I}_{\bar{n}}$ (includes cointegrated VAR(1))

$$\mathbb{E}\{\mathbf{X}_t | \mathbf{x}_0\} \stackrel{\mathbf{b} \approx \mathbb{I}_{\bar{n}}}{\approx} \mathbf{x}_0 + t\mathbf{a}$$

- Linear forecast error if $\mathbf{b} \approx \mathbb{I}_{\bar{n}}$ (includes cointegrated VAR(1))

□

$$\mathbb{C}v\{\mathbf{X}_t | \mathbf{x}_0\} \stackrel{\mathbf{b} \approx \mathbb{I}_{\bar{n}}}{\approx} t\sigma_{\hat{\varepsilon}}^2$$

$\approx$ random walk forecast ("square-root rule")

Linear state-space model
def.

Linear state-space model

Observation equation

$$\mathbf{X}_t = \boldsymbol{\mu} + \mathbf{c}\mathbf{H}_t + \hat{\varepsilon}_t \rightarrow \sim wvn(\mathbf{0}, \sigma_{\hat{\varepsilon}_t}^2)$$

Transition equation

$$\mathbf{H}_t = \mathbf{b}\mathbf{H}_{t-1} + \hat{\mathbf{U}}_t \rightarrow \text{hidden process}$$

$\dim(\mathbf{H}) = \text{small}$

$$\sim wvn(\mathbf{0}, \mathbb{I}_{\bar{k}}) \quad \bar{k} \text{ low}$$

$\equiv (\lambda_1, \dots, \lambda_{\bar{k}})'$
eigenvalues of $\mathbf{b}$

Cov. stationary

$$|eigval(\boldsymbol{\lambda})| < 1$$

Dynamic systematic idiosyncratic

$$\mathbb{C}v\{\hat{\varepsilon}_t\} = \text{Diag}(\mathbf{d})$$

Static systematic idiosyncratic

$$\mathbf{b} \equiv \mathbf{0}$$

Linear state-space model: main features

• is stationary

Forecast

Linear state-space model	
Observation equation	$X_t = \mu + cH_t + \varepsilon_t$
Transition equation	$H_t = bH_{t-1} + \dot{U}_t$

- Linear forecast $\equiv \{x_t, \dots, x_0\}$
 $\downarrow$
 $\mathbb{E}\{X_u | i_t\} = \mu + c\mathbb{E}\{H_u | i_t\}$

- Matrix forecast error

$$\mathbb{C}v\{X_u | i_t\} = c\mathbb{C}v\{H_u | i_t\}c' + \sigma_\varepsilon^2$$

$\Rightarrow$ we need $\mathbb{E}\{H_u | i_t\}, \mathbb{C}v\{H_u | i_t\}$

Forecast

Linear state-space model	
Observation equation	$X_t = \mu + cH_t + \varepsilon_t$
Transition equation	$H_t = bH_{t-1} + \dot{U}_t$

- Linear forecast hidden process $\rightarrow$ forecast hidden factors
 $\mathbb{E}\{H_u | i_t\} = b^{u-t} \hat{\mu}_t$

- Matrix forecast error hidden process

$$\mathbb{C}v\{H_u | i_t\} = b^{u-t} \hat{\sigma}_t^2 (b^{u-t})' + \gamma^2$$

$\uparrow$
 $\equiv \mathbb{C}v\{H_u | h_t\}$

$\Rightarrow$ we need $\hat{\mu}_t \equiv \mathbb{E}\{H_t | i_t\}, \hat{\sigma}_t^2 \equiv \mathbb{C}v\{H_t | i_t\}$

One more set of processes: Harmonic for seasonality

A univariate **harmonic process** X_t is defined as linear combinations of k trigonometric waves

$$X_t = \sqrt{2} \sum_{k=1}^k \sigma_k (\tilde{U}_{1,k} \cos(\omega_k t) + \tilde{U}_{2,k} \sin(\omega_k t)) \quad (43.156)$$

where

- the amplitudes $\tilde{U}_{k,i}$ are instance of a standardized weak white noise (43.5)

$$\tilde{U}_{k,i} \sim wwn(0, 1);$$

- the frequencies ω_k vary between 0 and π and ensure that the process is periodic, which means (RED)

$$\omega_k \in [0, \pi] \cap \{2\pi q\}_{q \in \mathbb{Q}},$$

where the restriction to $[0, \pi]$ is made without loss of generality (RED).

An equivalent way to introduce harmonic processes (43.156) is the below (RED)

$$X_t = 2 \sum_{k=1}^k \sigma_k \cos(\omega_k t + \tilde{\Phi}_k) \quad (43.159)$$

where $\tilde{\Phi}_k$ is uniform weak white noise

$$\tilde{\Phi}_k \sim Unif([- \pi, \pi]).$$

(43.160)

UNIT 3: Covariance stationary theory

• "all of econometrics"

$$\sigma^2 \equiv e \begin{pmatrix} \lambda & \dots & 0 \\ & \ddots & \vdots \\ 0 & \dots & \lambda \end{pmatrix} e' \rightarrow \sigma \equiv e d(\lambda) e'$$

$$\sigma^2 = SS'$$

special cases of sq. root
fundamental

RICATI $S \equiv \sigma \rightarrow$ so far

SPECTRAL $S \equiv e d(\sqrt{\lambda})$

CHOLESKY $S \equiv \Psi \Delta$

adv. econometrics

• discrete time, zero mean cov. stationary process

$$X_t \stackrel{mc}{\sim} K_x(\cdot)$$

We discuss in detail three fundamental results

- 1) The Wold decomposition assesses that any covariance stationary process (41.73)-(41.74) can be written as the sum of a linear combination, with matrix coefficients, of present and past instances of a weak white noise, and a fully predictable residual.
- 2) Spectral analysis and the Cramer decomposition assess that any covariance stationary process (41.73)-(41.74) is a superposition of uncorrelated harmonics (43.196), whose covariances are given by the Fourier transform of the autocovariance (41.74). It allows to better detect cyclic/seasonal patterns in covariance stationary processes.
- 3) The Wiener-Kolmogorov filter assesses the linear forecast (41.58) of $\mathbf{X}_{t+\Delta t}$ given the whole present, and past history, of another covariance stationary process $\mathbf{Z}_t$.

The main results presented in this chapter are summarized in the below Table 44.1.

Theory

- can be var one, linear-state, harmonic, VARMA, etc. GENERALIZES

stochastic process

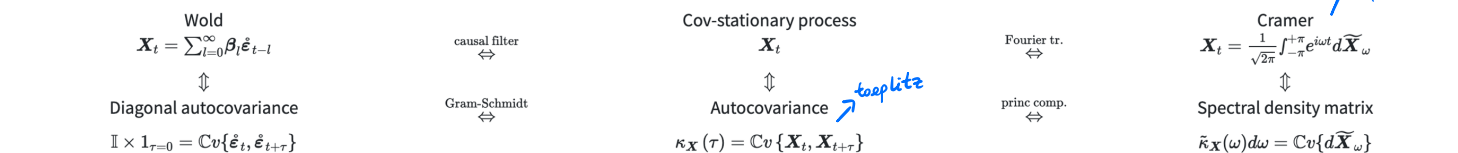


Table 44.1: Main results for covariance stationary processes

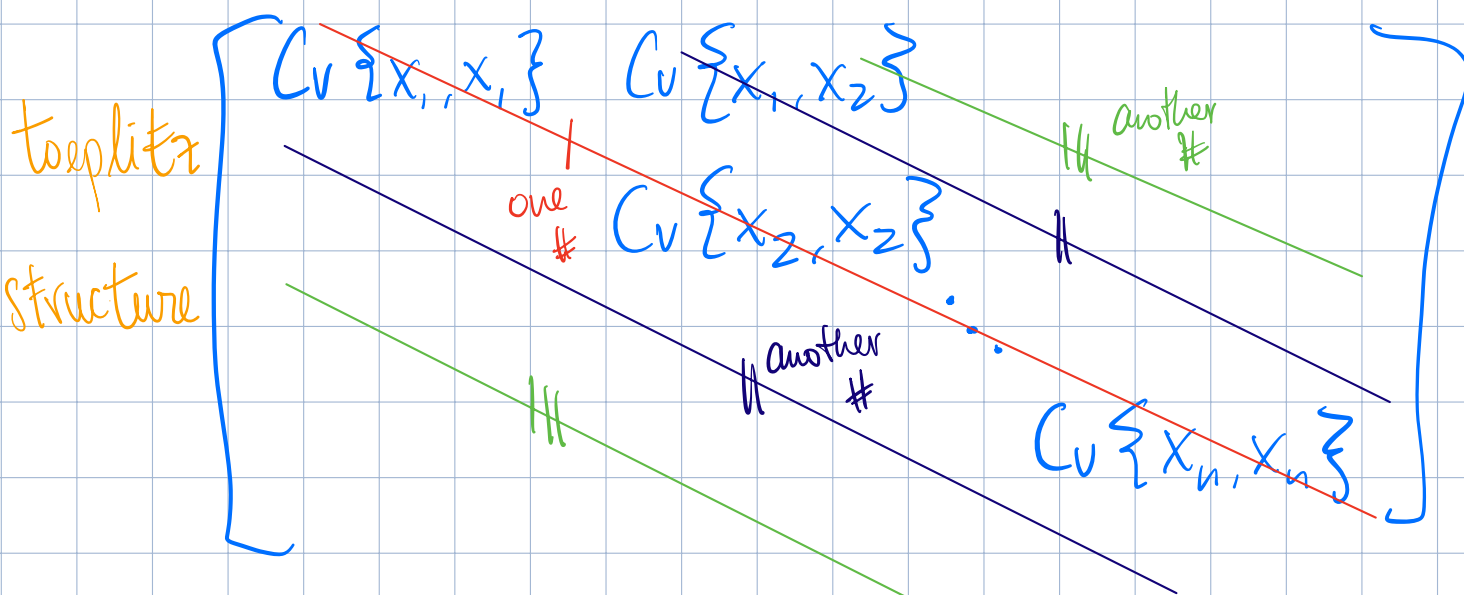
The Wold decomposition generalizes from Cholesky decomposition (48.497) of the joint covariance matrix of the variables $\mathbf{X}_{-l}, \dots, \mathbf{X}_l$ and the associated backward Gram-Schmidt orthonormalization algorithm (48.518).

Spectral analysis and the Cramer decomposition generalize from the asymptotic principal component decomposition of the joint covariance matrix of the variables $\mathbf{X}_{-l}, \dots, \mathbf{X}_l$

The Wiener-Kolmogorov filter express the linear forecast $\mathbb{E}\{\mathbf{X}_{t+\Delta t} \mid \mathbf{z}_t, \mathbf{z}_{t-1}, \dots\}$ (41.58) as a linear combination of infinite observations $\mathbf{z}_t, \mathbf{z}_{t-1}, \dots$, that we denote in compact form as

$$\mathbb{E}\{\mathbf{X}_{t+\Delta t} \mid \mathbf{z}_t, \mathbf{z}_{t-1}, \dots\} = \beta_{\mathbf{X} \mid \mathbf{Z}}^{(\Delta t)}(0) \mathbf{z}_t + \beta_{\mathbf{X} \mid \mathbf{Z}}^{(\Delta t)}(1) \mathbf{z}_{t-1} + \dots \equiv [\beta_{\mathbf{X} \mid \mathbf{Z}}^{(\Delta t)} * \mathbf{z}]_t, \quad (44.2)$$

where the coefficients $\beta_{\mathbf{X} \mid \mathbf{Z}}^{(\Delta t)}(l)$ are connected to the Wold decomposition and the spectral decomposition of both processes.



Any $n \times n$ matrix A of the form

$$A = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & \cdots & \cdots & a_{-(n-1)} \\ a_1 & a_0 & a_{-1} & \ddots & & \vdots \\ a_2 & a_1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & a_{-1} & a_{-2} \\ \vdots & & \ddots & a_1 & a_0 & a_{-1} \\ a_{n-1} & \cdots & \cdots & a_2 & a_1 & a_0 \end{bmatrix}$$

is a Toeplitz matrix. If the i, j element of A is denoted $A_{i,j}$ then we have

$$A_{i,j} = A_{i+1,j+1} = a_{i-j}.$$

Wold representation intuition

Intuition - Finite time points

Goal: represent X_t as causal linear transformation of uncorrelated shocks

Autocovariance matrix: symmetric, positive definite

$$\text{Cov}\{X_{-\bar{t} \rightarrow \bar{t}}\} = \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix} \times \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix}^T$$

↑
Cholesky decomposition

Standardized weak white noise

$$\hat{U}_{-\bar{t} \rightarrow \bar{t}} \equiv \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix}^{-1} X_{-\bar{t} \rightarrow \bar{t}}$$

$\hookrightarrow p_V$

Vector of univariate cov. stationary process $X_t \sim \mathbb{C}_V \text{ } K_X(\cdot)$

$$X_{-\bar{t} \rightarrow \bar{t}}$$

Process as linear combination of white noise

$$X_{-\bar{t} \rightarrow \bar{t}} = \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix} \hat{U}_{-\bar{t} \rightarrow \bar{t}}$$

$\Downarrow$

Solution: linear transformation

$$X_t = \sum_{s \leq t} \psi_{t,s} \hat{U}_s$$

ψ lower triangular

$\Rightarrow$ causal linear combination of white noise

of past shocks

Autocovariance matrix: symmetric, positive definite

$$\text{Cov}\{X_{-\bar{t} \rightarrow \bar{t}}\} = \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix} \times \begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix}^T$$

$\bar{t} \rightarrow \infty \quad \Downarrow \quad \text{Cov}\{X_{-\bar{t} \rightarrow \bar{t}}\} \text{ Toeplitz}$

$$\begin{pmatrix} \psi_{-\bar{t}, -\bar{t}} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_{0, -\bar{t}} & \vdots & \psi_{0, 0} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{\bar{t}, -\bar{t}} & \dots & \psi_{\bar{t}, 0} & \dots & \psi_{\bar{t}, \bar{t}} \end{pmatrix} \xrightarrow[\approx]{\bar{t} \rightarrow \infty} \begin{pmatrix} \psi_0 & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \psi_l & \vdots & \psi_0 & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{2l} & \dots & \psi_l & \dots & \psi_0 \end{pmatrix}$$

Solution: linear time invariant filter

$$X_t \stackrel{\bar{t} \rightarrow \infty}{=} \sum_{l \geq 0} \psi(l) \hat{U}_{t-l} = [\psi * \hat{U}]_t$$

ψ lower triangular, Toeplitz $\Rightarrow$ causal linear time invariant filter of white noise

Time Series and Sequential Decisions > Covariance stationary theory > Wold representation
Formal statement

Formal statement

Wold decomposition theorem

$$X_t = X_t^{lr} + X_t^{ld}$$

uncorrelated

In general cov. stationary X_t is the sum of two covariance stationary processes

• Linearly regular component

$$X_t^{lr} = \sum_{l=0}^{\infty} \psi_X(l) \hat{U}_{t-l} = [\psi_X * \hat{U}]_t$$

$\uparrow$
 $\hat{U}_t \sim wwn(0, \mathbb{I}_{\bar{K}})$

*through
Graham-
Schmidt*

- **Linearly deterministic component**

$$\mathbf{X}_t^{ld} = \mathbb{E}\{\mathbf{X}_t^{ld} | \mathbf{I}_{t-1}\}$$

can be exactly predicted

Two components are uncorrelated

$$\text{Cov}\{\mathbf{X}_t^{lr}, \mathbf{X}_s^{ld}\} = \mathbf{0}$$

Relationship with spectral analysis

Wold decomposition theorem

-

Cramer's decomposition

$$\mathbf{X}_t = \mathbf{X}_t^{lr} + \mathbf{X}_t^{ld} = \mathbf{X}_t^{ac} + \mathbf{X}_t^{jump}$$

In most situations

- **Linearly regular component**

$$\mathbf{X}_t^{lr} = \sum_{l=0}^{\infty} \psi_{\mathbf{X}}(l) \dot{\mathbf{U}}_{t-l} = [\psi_{\mathbf{X}} * \dot{\mathbf{U}}]_t \sim_{\text{Cv}} \tilde{\kappa}_{\mathbf{X}}^{ac}(\omega)$$

$$\tilde{\kappa}_{\mathbf{X}}^{ac}(\omega) = \tilde{\psi}_{\mathbf{X}}(\omega) \tilde{\psi}_{\mathbf{X}}(\omega)^H$$

- ! General technique to compute Wold coefficients $\psi_{\mathbf{X}}(l)$, because Gram-Schmidt cannot be applied in practice

o Finite time points:

- triangularity in Cholesky $\Rightarrow$ Wold makes it only a linear combo of past entries
- in spectral, eigenvector matrix has special cols.

profile of cols looks like waves of diff. shapes.

Filtering

- analyzes MCF linear transformations

Filtered process - finite case

Finite instances of covariance stationary process

$$\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}} \equiv \mathbf{a} \times \mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}$$

Autocovariance matrix: symmetric, positive definite

$$\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \mathbf{a} \times \mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} \times \mathbf{a}'$$

↓ Toeplitz

$$\mathbf{a} = \begin{pmatrix} a(0) & \dots & a(\bar{t}) & \dots & a(2\bar{t}) \\ \vdots & & \ddots & & \vdots \\ a(\bar{t}) & & & & a(\bar{t}) \\ \vdots & & & \ddots & \vdots \\ a(2\bar{t}) & \dots & a(\bar{t}) & \dots & a(0) \end{pmatrix}$$

↳ we wanted in Cholesky root s.t. it becomes a convolution

Filtered process

$$Y_t = \sum_{l=t-\bar{t}}^{t+\bar{t}} a(l) X_{t-l}$$

Filtered process - finite case

Finite instances of covariance stationary process

$$\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}} \equiv \mathbf{a} \times \mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}$$

Autocovariance matrix: symmetric, positive definite

$$\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \mathbf{a} \times \mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} \times \mathbf{a}'$$

↓ Toeplitz

$$\mathbf{a} = \begin{pmatrix} a(0) & \dots & a(\bar{t}) & \dots & a(2\bar{t}) \\ \vdots & & \ddots & & \vdots \\ a(\bar{t}) & & & & a(\bar{t}) \\ \vdots & & & \ddots & \vdots \\ a(2\bar{t}) & \dots & a(\bar{t}) & \dots & a(0) \end{pmatrix} \quad \mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \begin{pmatrix} \kappa_X(0) & \dots & \kappa_X(\bar{t}) & \dots & \kappa_X(2\bar{t}) \\ \vdots & & \ddots & & \vdots \\ \kappa_X(\bar{t}) & & & & \kappa_X(\bar{t}) \\ \vdots & & & \ddots & \vdots \\ \kappa_X(2\bar{t}) & \dots & \kappa_X(\bar{t}) & \dots & \kappa_X(0) \end{pmatrix}$$

Filtered process

$$Y_t = \sum_{l=t-\bar{t}}^{t+\bar{t}} a(l) X_{t-l}$$

Filtered autocovariance

$$\kappa_Y(t+\tau, t) = \sum_{l=-\bar{t}+t+\tau}^{\bar{t}+t+\tau} \sum_{m=-\bar{t}-t}^{\bar{t}-t} a(l) \kappa_X(\tau-l-m) a(m)$$

Filtered process - infinite case

Finite instances of covariance stationary process

$$\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}} \equiv \mathbf{a} \times \mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}$$

Autocovariance matrix: symmetric, positive definite

$$\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \mathbf{a} \times \mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} \times \mathbf{a}'$$

! $\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\}$ is Toeplitz $\Downarrow$ Toeplitz, $\bar{t} \rightarrow \infty$

Covariance stationary filtered process

Filtered process

$$\mathbf{Y}_t \xrightarrow{\bar{t} \rightarrow +\infty} \sum_{l=-\infty}^{+\infty} a(l) X_{t-l} \equiv [a * X]_t$$

Filtered autocovariance

only depends on lag τ

$$\kappa_Y(t + \tau, t) \xrightarrow{\bar{t} \rightarrow +\infty} \kappa_Y(\tau) \equiv [a * \kappa_X * a](\tau)$$

Filtered process - infinite case

Finite instances of covariance stationary process

$$\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}} \equiv \mathbf{a} \times \mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}$$

Autocovariance matrix: symmetric, positive definite

$$\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \mathbf{a} \times \mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} \times \mathbf{a}'$$

Spectral decomposition

diff spectra

same ee'

$$\mathbb{C}v\{\mathbf{X}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \frac{\Delta\omega}{2\pi} \boldsymbol{\eta}_X \times \text{Diag}(\boldsymbol{\lambda}_X) \times \boldsymbol{\eta}_X'$$

$$\mathbb{C}v\{\mathbf{Y}_{-\bar{t} \rightsquigarrow \bar{t}}\} = \frac{\Delta\omega}{2\pi} \boldsymbol{\eta}_Y \times \text{Diag}(\boldsymbol{\lambda}_Y) \times \boldsymbol{\eta}_Y'$$

$$\mathbf{a} = \frac{\Delta\omega}{2\pi} \boldsymbol{\eta}_a \times \text{Diag}(\boldsymbol{\lambda}_a) \times \boldsymbol{\eta}_a'$$

$\Downarrow$ Toeplitz, $\bar{t} \rightarrow \infty$

$$\boldsymbol{\eta}_a \approx \boldsymbol{\eta}_Y \approx \boldsymbol{\eta}_X \approx \boldsymbol{\eta}$$

$$\boldsymbol{\eta} \text{Diag}(\boldsymbol{\lambda}_Y) \boldsymbol{\eta}' \xrightarrow{\bar{t} \rightarrow +\infty} \boldsymbol{\eta} \text{Diag}(\boldsymbol{\lambda}_a) \text{Diag}(\boldsymbol{\lambda}_X) \text{Diag}(\boldsymbol{\lambda}_a) \boldsymbol{\eta}'$$

external eigenvectors

Generate image

Simplify

Define

Ask AI Assistant

Linear filter

$$\mathbf{X}_t \stackrel{mc}{\sim} \kappa_{\mathbf{X}}(\cdot, \cdot)$$

Linear time invariant filter

$$\mathcal{A}[\mathbf{X}]_t \equiv \sum_{l=-\infty}^{+\infty} a(l) \mathbf{X}_{t-l} = [a * \mathbf{X}]_t$$

Impulse response function

$$a(l) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i\omega l} \tilde{a}(\omega) d\omega$$

Frequency response function

$$\tilde{a}(\omega) \equiv \sum_{l \in \mathbb{Z}} e^{-i\omega l} a(l)$$



Fourier transform

Frequency response function of the LTI filter

Spectral density of filtered processes

If $\mathbf{X}_t$ is covariance stationary, $[a * \mathbf{X}]_t$ is covariance stationary

$$\mathbf{X}_t \stackrel{mc}{\sim} \kappa_{\mathbf{X}}(\cdot) \xRightarrow{a \text{ LTI}} [a * \mathbf{X}]_t \stackrel{mc}{\sim} \kappa_{a * \mathbf{X}}(\cdot) \Leftrightarrow \tilde{\kappa}_{a * \mathbf{X}}(\cdot)$$

Linear time invariant filter

$$\mathcal{A}[\mathbf{X}]_t \equiv \sum_{l=-\infty}^{+\infty} a(l) \mathbf{X}_{t-l} = [a * \mathbf{X}]_t$$

Impulse response function

$$a(l) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i\omega l} \tilde{a}(\omega) d\omega$$

Frequency response function

$$\tilde{a}(\omega) \equiv \sum_{l \in \mathbb{Z}} e^{-i\omega l} a(l)$$

Autocovariance function of filtered process

$$\tilde{\kappa}_{a * \mathbf{X}}(\tau) = [a * \kappa_{\mathbf{X}} * a^{-}]^{\mathcal{I}}(\tau)$$

Spectral density of filtered process

$$\tilde{\kappa}_{a * \mathbf{X}}(\omega) = \tilde{a}(\omega) \times \tilde{\kappa}_{\mathbf{X}}(\omega) \times \tilde{a}(\omega)^H$$

UNIT 4: Probabilistic Stochastic Models

- MCF probabilistic state based models

observation eq + transition eq
Markov process

- special case to go discrete to discrete
eg. credit rating $\rightarrow$ (default, no default)

Markov chains

A **time-homogeneous Markov chain** is a Markov chain (45.51), where the conditional probability mass function (58a.60) are constant **transition probabilities** from the generic value $X_t = r$ to the generic next-step value $X_{t+1} = c$

$$p_{r \rightarrow c} \equiv \mathbb{P}\{X_{t+1} = c | X_t = r\}. \quad (45.55)$$

Hence, in a time homogeneous Markov chain, the conditional distributions (45.51) are scenario-probability distributions (54.266), as follows

$$X_{t+1} | \{X_t = r\} \sim \{c, p_{r \rightarrow c}\}_{c=1}^{\bar{c}}, \quad (45.56)$$

for all $r = 1, \dots, \bar{c}$. Then the respective scenario-probability conditional cdf $F_X(\cdot | r)$ (54.304) reads

$$F_X(c | r) = \sum_{u \leq c} p_{r \rightarrow u}. \quad (45.57)$$

Example 45.11.

In Example 58a.27 we visualize the scenario-probability distribution $\{c, p_{r \rightarrow c}\}_{c=1}^{\bar{c}}$ (45.56) and the conditional cdf $F_X(c | r)$ (45.57) for the initial state r in the context of modeling credit rating transitions.

The transition probabilities (45.55) can be conveniently rearranged in a $\bar{c} \times \bar{c}$ **time-homogeneous transition matrix**

$$P \equiv \begin{pmatrix} p_{1 \rightarrow 1} & \cdots & p_{1 \rightarrow c} & \cdots & p_{1 \rightarrow \bar{c}} \\ \vdots & & \vdots & & \vdots \\ p_{r \rightarrow 1} & \cdots & p_{r \rightarrow c} & \cdots & p_{r \rightarrow \bar{c}} \\ \vdots & & \vdots & & \vdots \\ p_{\bar{c} \rightarrow 1} & \cdots & p_{\bar{c} \rightarrow c} & \cdots & p_{\bar{c} \rightarrow \bar{c}} \end{pmatrix}, \quad \rightarrow \text{each row is a PMF going from one state to another, in steps} \quad (45.58)$$

where the generic conditional transition probability $p_{r \rightarrow c}$ (45.55) corresponds to the r -th row and c -th column of the transition matrix (45.58).

GARCH

- models vol clustering

Skipped unit 5

UNIT 6: Sequential decisions theory

- deals w/ dynamic causality (like intervention through time)
- multi-period decisions are constrained by past inputs (triangular)
↳ get dynamic sequence decision.

decision theory ^{past only} → multi-period control
linear quadratic Gaussian control

Consider a POMDP problem (47.56), as described by the states (47.32), inputs (47.33), and actions (47.34). Assume that the system is a (generalized) linear state-space model (Table 43.5) described by the following equations.

- A (generalized) linear transition equation (43.82) for the hidden state process S_t (47.32), driven by the actions A_t (47.34)

$$S_t = bS_{t-1} + gA_{t-1} + \tilde{U}_t, \quad \text{state depends on actions} \quad (47.83)$$

where b and g are matrices of suitable dimensions and where A_t is typically selected through some policy π_t (47.36). Here we assume that all the eigenvalues of the matrix b (47.88) are smaller than one in magnitude

$$|\text{eigval}(b)| < 1, \quad (47.84)$$

where the absolute value is meant entry by entry and the inequalities component by component. The requirement (47.84) is the same condition that guarantees covariance stationarity for the ensuing linear state-space model (43.93).

- A linear observation equation (43.81) for the input process X_t (47.33)

$$X_t = cS_t + \tilde{\epsilon}_t. \quad (47.85)$$

The random variables $\tilde{U}_t$ (47.83) and $\tilde{\epsilon}_t$ (47.85) are uncorrelated Gaussian white noises (42.3) with zero mean and constant covariance

$$\begin{pmatrix} \tilde{U}_t \\ \tilde{\epsilon}_t \end{pmatrix} \sim N\left(\begin{pmatrix} 0_n \\ 0_m \end{pmatrix}, \begin{pmatrix} \sigma_{\tilde{U}}^2 & 0_{\tilde{m} \times \tilde{n}} \\ 0_{n \times m} & \sigma_{\tilde{\epsilon}}^2 \end{pmatrix}\right). \quad (47.86)$$

Moreover, the hidden process (47.83) at time $t = 0$ is set to have an arbitrary Gaussian distribution (54.1)

$$S_0 \sim N(\mu_{S_0}, \sigma_{S_0}^2). \quad (47.87)$$

uncorrelated with the driving Gaussian white noise $\tilde{\epsilon}_t$, $\tilde{U}_t$ (47.86) at all times.

Note that the state transition function (47.83) induces a Gaussian transition probability function (47.46) (47)

$$S_{t+1}|s_t, a_t \sim f^N(s_{t+1}|bs_t + ga_t, \sigma_{\tilde{U}}^2). \quad (47.88)$$

Moreover, the observation function (47.85) induces a Gaussian observation probability function (47.47) independent of the actions (47.34) (47)

$$X_t|s_t, a_{t-1} \sim f^N(x_t|cs_t, \sigma_{\tilde{\epsilon}}^2). \quad (47.89)$$

Moreover, we assume that the reward (47.54) be quadratic in the joint states (47.32) and actions (47.34)

$$\bar{w}(s_t, a_t) \equiv - \begin{pmatrix} s_t^{\top} & a_t^{\top} \end{pmatrix} \begin{pmatrix} q^2 & p \\ p^{\top} & r^2 \end{pmatrix} \begin{pmatrix} s_t \\ a_t \end{pmatrix}, \tag{47.93}$$

where q^2 and r^2 are positive-semidefinite matrices (48.150) of suitable dimensions. Here, p is a matrix of suitable dimensions on which we apply some mild constraints to ensure that the reward be always negative (48). It is also possible to account for an additional linear term in the reward (47.93) by including the constant scalar 1 among the states (48).

The goal (47.55) ensuing from the reward (47.93) here reads

$$G_t \equiv - \sum_{u \geq t} \gamma^{u-t} \begin{pmatrix} S_u^{\top} & A_u^{\top} \end{pmatrix} \begin{pmatrix} q^2 & p \\ p^{\top} & r^2 \end{pmatrix} \begin{pmatrix} S_u \\ A_u \end{pmatrix}. \tag{47.94}$$

Alert 47.8.

Many real applications necessitate a finite temporal horizon τ , as opposed to the infinite one we consider in the goal (47.94).

In such cases, instead of G_t (47.94) we consider the **discounted cumulative reward** $G_{t,\tau}$, defined as

↳ cares more about recent past than older past

$$G_{t,\tau} \equiv - \sum_{t \leq u \leq t+\tau} \gamma^{u-t} \begin{pmatrix} S_u^{\top} & A_u^{\top} \end{pmatrix} \begin{pmatrix} q^2 & p \\ p^{\top} & r^2 \end{pmatrix} \begin{pmatrix} S_u \\ A_u \end{pmatrix}. \tag{47.95}$$

Here we neglect the corrections introduced by the horizon finiteness (47.95), as $G_{t,\tau}$ is well approximated by the goal (47.94) when τ is large enough due to the exponential decay induced by γ (48).

$$G_{t,\tau} \approx G_t. \tag{47.96}$$

Finally, if we consider policies in the form (47.36), the **linear-quadratic Gaussian control** problem is the joint optimization of the expected goal G_0 (47.94) over the policy sequences $\bar{\pi}_0$ (47.37) which returns the linear-quadratic Gaussian controls $\bar{\pi}_0^*$

$$\bar{\pi}_0^* \equiv \operatorname{argmax}_{\bar{\pi}_0} \mathbb{E} \left\{ - \sum_{t \in \mathbb{N}} \gamma^t \begin{pmatrix} S_t^{\top} & A_t^{\top} \end{pmatrix} \begin{pmatrix} q^2 & p \\ p^{\top} & r^2 \end{pmatrix} \begin{pmatrix} S_t \\ A_t \end{pmatrix} \mid \operatorname{do}(\bar{\pi}_0) \right\}. \tag{47.97}$$

↳ computable

Wealth Portfolio Optimization

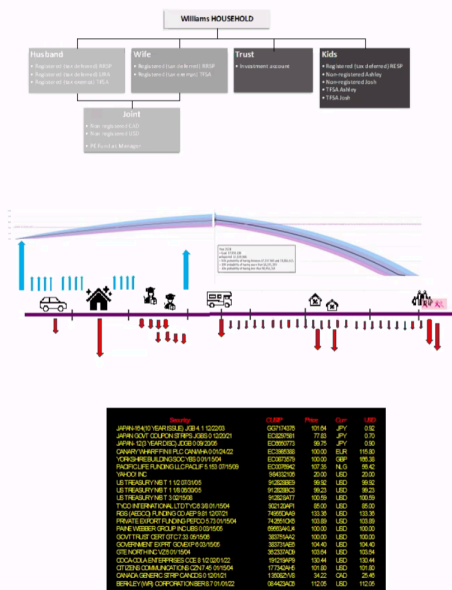
Efficient Portfolio Manufacturing and Client Management at Scale!



Objective: provide **personalized**, and **high-touch** portfolio investment programs for 100s or 1,000s of clients in an efficient manner

- Understand the entire book of business
 - Pro-actively (instead of reactively) address which clients
 - Require more attention
 - Can take advantage of an opportunity
 - Are more affected by a given market event...

Portfolio Optimization in Wealth Management



“Optimal” Household portfolio strategy

- **Maximize multi-criteria (utility) function:** multiple goals over time
 - Inflows and outflows (consumption), terminal wealth
- Long horizon (covering accumulation and decumulation)
- Takes best advantage of investment universe available (“efficient portfolios”)
- Accounting for taxes, risk profiles, stochastic markets
- Complex constraints: consistent with multiple client risk/investment preferences, liquidity, compliance/regulatory (tax)

Applied Financial Finance

Dilip Madan

- Focused on not losing reality during quantitative exercises. Argue to push objective functions (MCF, vols) aside, and how to replace them with a different set.
- Automated Investment Problem (AIP)
 - Quant investment strategies should not just say buy/sell but where and how much.
 -
- Portfolio Theory Constraints
- Nonlinear Expectation Solutions and their Issues
- The Finance Finance Solution
- Disciplined Saddle Point Programming
- Single Asset Applications
- Multi Asset Applications
- Conclusions